

FINAL (ATTRIBUTES/PARAMETERS)

Variable	Method	Class
Constant	No overriding	No inheritance

INITIALISATION

1) Default-Values ↓

2) Attribute Assignments

3) Initialisation block

4) Constructor

DEFAULT VALUES

Type	Default	Type	Default
boolean	false	char	'\u0000'
byte	0	short	0
int	0	long	0L
float	0.0f	double	0.0d

TYPES

Type	Size (bit)	From	To
byte	8	−128	127
short	16	−32'768	32'767
char	16	UTF-16 chars	
int	32	−2 ³¹	2 ³¹ − 1
long	64	−2 ⁶³	2 ⁶³ − 1
float	32	±1.4 · 10 ^{−45}	±3.4 · 10 ³⁸
double	64	±4.9 · 10 ^{−324}	±1.7 · 10 ³⁰⁸

// short * int ⇒ int

// float + int ⇒ float

// int / double ⇒ double

// int + long * float ⇒ float

// 0.0 / 0.f ⇒ NaN

long l = 1L;

long ll = 0b1L;

float f = 0.0f;

double d = 0.0d;

12 = '.'; // implicit int/char conversion

0.1 + 0.1 ≠ 0.2; // true

5/2 = 2; // true, int div truncates to 0

NaN = NaN; // false

Integer.MAX_VALUE + 1 = Integer.MIN_VALUE;

1 / 0.0 = Double.POSITIVE_INFINITY; // true

var ints = new ArrayList<Integer>();

int[] jnts = new int[69];

int[][] matrix = new int[6][9];

if (obj instanceof ArrayList<Integer>)

((ArrayList<Integer>)obj).add(2);

public List<String> method()

BiFunction<Integer, String, List<String>> fn) {

return fn.apply(5, "FooBar");

}

IMPLICIT CASTING

No information loss int→float, to larger type int→long

Sub→Super is implicit, Super→Sub ClassCastException

// explicit casting

float f = (float) 1;

// type conversion

Integer.parseInt("2");

Float.parseFloat("2.0");

WIDENING PRIMITIVE CONVERSION

– If either operand is double, the other is converted to double

– Otherwise, if either is float, the other is converted to float

– Otherwise, if either is long, the other is converted to long

– Otherwise, both operands are converted to type int

VARIABLE ARGS

static int sum(int... numbers) {

int sum = 0;

for (int i = 0; i < numbers.length; i++)

sum += numbers[i];

return sum;

}

MISC

int[] intarr = new int[] {1, 2, 3, 4, 5};

int[] sub = Arrays.copyOfRange(intarr, 1, 3); // 2,3

var intlist = new ArrayList<Integer>();

intarr.length;

intlist.size();

// Multiply first to not lose precision

int percent = (int)((filled * 100) / capacity);

obj.clone();

Math.min(x, y);

Math.max(x, y);

Rekapitulation: Primitive Datentypen

Konversionen

double

float

long

int

short

byte

char

explicit

implizit (mit evtl. Genauigkeitsverlust)

Sonstige Richtungen implizit

EQUALITY

s.equals(sOther); // Strings / Objects

Arrays.equals(a1, a2); // arrays

Arrays.deepEquals(a1, a2); // nested arrays

class Student extends Person {

@Override

public boolean equals(Object obj) {

if (obj == null) return false;

if (getClass() != obj.getClass()) return false;

if (!super.equals(obj)) return false;

Student other = (Student) obj;

return getNumber() == other.getNumber();

}

STRINGS

String multiline = ""

Hello, "world"

"";

"a:b:c".split(":",2).length == 2; // true

String str = Integer.toString(123456789);

str.length(); // 9

str.charAt(1); // 2

str.toUpperCase();

str.toLowerCase();

str.trim();

str.substring(1, 3); // 2,3

STRING POOLING

String first = "hello", second = "hello";

System.out.println(first == second); // true

String third = new String("hello");

String fourth = new String("hello");

System.out.println(third == fourth); // false

System.out.println(third.equals(fourth)); // true

String a = "A", b = "B", ab = "AB";

System.out.println(a + b == ab); // false

final String d = "D", e = "E", de = "DE";

System.out.println(d + e == de); // true

VISIBILITY

public	all classes
protected	package + sub-classes
private	only self
(none)	all classes in same package

PACKAGES

p1.sub won't be automatically imported in p1.

Package name collisions: first gets imported.

package p1;

public class A { }

package p1.sub;

public class E { }

package p2;

import p1.sub.E;

public class C {

E e = new E();

}

package p1; // public class A { }

package p2; // public class A { }

// OK

import p1.A;

import p2.*;

// reference to A is ambiguous

import p1.*;

import p2.*;

// sin, PI

import static java.lang.Math.*;

MODULES

./foo/module-info.java

module foo.bar.baz {

exports com.my.package.foo;

}

./main/module-info.java

module main.module {

requires com.my.package.foo;

}

ENUMS

public enum Weekday {

MONDAY(true),

TUESDAY(true),

WEDNESDAY(true),

THURSDAY(true),

FRIDAY(true),

SATURDAY(false),

SUNDAY(false);

private boolean workDay;

Weekday(boolean workDay) { // private constructor

this.workDay = workDay;

}

public boolean isWorkDay() {

return workDay;

}

switch (wd) {

case Weekday.MONDAY:

case Weekday.TUESDAY:

System.out.println("First two");

break;

default:

System.out.println("Rest");

}

public enum Level {

LOW,

MEDIUM,

HIGH

}

SWITCH

switch (x) {

case 'a':

System.out.println("1");

break;

default:

System.out.println("2");

}

int v = switch (x) {

case 'a' -> 1;

default -> 2;

}

SHADOWING

Variables with same names in same scope

HIDING

Variables with same names in different classes

Gets statically chosen by compiler

super.description = ((Vehicle)this).description

super.super // doesn't exist, use v

((SuperSuperClass)this).variable

// for multiple parent interfaces v

ParentInterface1.super.defaultMethod();

// ^ executes defaultMethod on ParentInterface1

ParentInterface2.super.defaultMethod();

OVERLOADING

Methods with same names but different parameters

Gets statically chosen by compiler

void print(int i, double j) {} // 1

void print(double i, int j) {} // 2

void print(double i, double j) {} // 3

print(1.0, 2.0); // 3

print(1,2); // error: reference to print is ambiguous

print(1.0, 2); // 2

print(2.0, (double) 2); // 3

OVERRIDING

Methods with same names and signatures

Dynamically chosen (Dynamic dispatch / Virtual call)

Error: Cannot override the final method...

Error: Cannot be subclass of final class...

class Fruit {

void eat(Fruit f) { System.out.println("1"); }

}

class Apple extends Fruit {

void eat(Fruit f) { System.out.println("2"); }

void eat(Apple a) { System.out.println("3"); }

}

Apple a = new Apple();

Fruit fa = new Apple();

Fruit f = new Fruit();

a.eat(fa); // 2

a.eat(a); // 3

fa.eat(a); // 2

fa.eat(fa); // 2

f.eat(fa); // 1

f.eat(a); // 1

((Fruit) a).eat(fa); // 3

((Apple) fa).eat(a); // 2

((Apple) f).eat(a); // ClassCastException

ABSTRACT CLASSES

public abstract class Vehicle {

private int speed;

public abstract void drive();

public void accelerate(int acc) {

this.speed += acc;

}

}

public class Car extends Vehicle {

@Override

public void drive() { }

@Override

public void accelerate (int acc) { }

}

public class Qwer {

public void print() {

System.out.println("1");

}

}

public class Asdf extends Qwer {

@Override

public void print() {

System.out.println("2");

}

public void dostuff () { }

}

var x = new Asdf();

x.print(); // 2

((Qwer) x).print(); // 2

((Qwer) x).dostuff(); // cannot find symbol

Static Type: According to var declaration at compiletime

Dynamic Type: Type of the instance at runtime

INTERFACES

Cannot have Attributes

interface RoadV {

int MAX_SPEED = 120;

void drive();

}

interface WaterV {

int MAX_SPEED = 80;

void drive();

}

class AmphibianMobile implements RoadV, WaterV {

@Override // because ambiguous

public void drive() {

println(RoadV.MAX_SPEED); // MAX_SPEED ambiguous

}

}

interface RoadV { String getModel(); }

interface WaterV { int getModel(); }

// Error, because of different return types

class AmphibianMobile implements RoadV, WaterV { }

INTERFACES DEFAULT METHODS

interface Vehicle {

default void printModel() {

System.out.println("Undefined vehicle model");

}

}

ANONYMOUS CLASSES

var v = new RoadV() {

@Override

public void drive() {

System.out.println("Anon");

}

}

INHERITANCE

public class Vehicle {

private int speed;

public Vehicle(int speed) {

this.speed = speed;

}

}

public class Car extends Vehicle {

private int doors;

public Car(int speed, int doors) {

super(speed);

this.doors = doors;

}

}

Car c = new Car(); // Points to Car

Vehicle v = new Car(); // Points to Car

Object o = new Car(); // Points to Car

// ^static ^dynamic

Car c = (Car) new Vehicle(); // ClassCastException

MORE INHERITANCE

public class Qwer {

public void print() {

System.out.println("1");

}

}

public class Asdf extends Qwer {

@Override

public void print() {

System.out.println("2");

}

public void dostuff () { }

}

```
ITERATORS
Iterator<String> it = stringList.iterator();
while (it.hasNext()) {
    String s = it.next();
    System.out.println(s);
}
```

Mutating Collection while iterating over it: ConcurrentModificationException

EXCEPTIONS	
Error	Exception
Critical, don't handle	Runtime, handleable
OutOfMemoryError, StackOverflowError, AssertionError	IOException
Checked	Unchecked
Must be handled (or throws-declaration)	Not necessary
Checked by compiler	Compiler doesn't check
Exception, not RuntimeException	RuntimeException, Error

Child Exception gets caught in catch clause with parent class

```
void test() throws ExceptionA, ExceptionB {
    String c = cliP("asdf");
    throw new ExceptionB("wack");
}

// finally ALWAYS executes, even on unhandled Exc.
try {
    test();
} catch (ExceptionA | ExceptionB e) {
    // ...
} finally { }

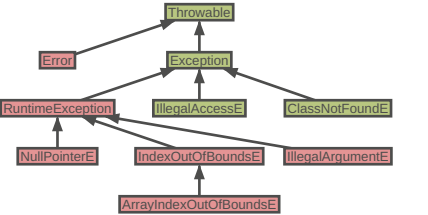
try { ... } catch(NullPointerException e) {
    throw e; // →leaves blocks→
} catch (Exception e) {
    // above e won't get caught!
} finally {
    // will still get executed
}

1 / 0; // ArithmeticException div by zero

String s = "";
s = null;
s.toUpperCase(); // NullPointerException

int[] arr = new int[] {1, 2, 3};
int elem = arr[8]; // ArrayIndexOutOfBoundsException
```

UncheckedChecked



IMPORTANT STUFF

- Hashing should be added to equals fn's for strict equality
- Check if input == null
- Check if array.length == 0
- IllegalArgumentException("reason")
- try/catch finally block always executes

```
IO
try (var fr = new FileReader("text.txt")) {
    int input = fr.read();
    while (input >= 0) {
        if (input == ';') { /* do something */ }
        input = fr.read();
    }
}

try (FileWriter writer = new FileWriter("out.txt",
    StandardCharsets.UTF_8, true)) { // append
    writer.write("weooo\n");
}

try {
    var input = new FileInputStream("text.txt");
    int i = input.read();
    while(i != -1) {
        System.out.print((char)i);
        i = input.read();
    }
    input.close();
} catch (Exception e) {
    e.printStackTrace();
}

try (BufferedReader reader = new BufferedReader(
    new FileReader("text.txt",
        StandardCharsets.UTF_8))) {
    String line;
    while ((line = reader.readLine()) != null) {
        System.out.println(line);
    }
}

try (
    FileReader reader = new FileReader("in.txt");
    FileWriter writer = new FileWriter("out.txt")
) {
    int i = input.read();
    while(i >= 0) {
        writer.write(i);
        i = input.read();
    }
}
```

TRY WITH

```
try (var output = new FileOutputStream("f.txt")) {
    output.write("Hello".getBytes());
} catch (IOException e) {
    System.out.println("Error writing file");
} finally {
    System.out.println("Done");
}
```

SERIALIZING

```
class X implements Serializable { }
// Serializing
try (var stream = new ObjectOutputStream(
    new FileOutputStream("s.bin"))) {
    stream.writeObject(new X());
}
// Deserializing
try (var stream = new ObjectInputStream(
    new FileInputStream("s.bin"))) {
    X x = (X) stream.readObject();
}
```

FUNCTION

```
public interface Function<T, R> {
    R apply(T t);

    @FunctionalInterface
    static <T> Function<T, T> identity();

    <V> Function<T, V> andThen(
        Function<? super R, ? extends V> after);

    <V> Function<V, R> compose(
        Function<? super V, ? extends T> before);
}
```

```
PREDICATE
public interface Predicate<T> {
    boolean test(T t);
}
```

```
static void removeAll(Collection<Person> collection,
    Predicate criterion) {
    var it = collection.iterator();
    while (it.hasNext())
        if (criterion.test(it.next()))
            it.remove();
}
```

COMPARABLE

```
public interface Comparable<T> {
    int compareTo(T obj);
}
```

```
var l = new ArrayList<Integer>(asList(3,2,4,5,1));
l.sort((a, b) -> a > b ? 1 : -1); // =
l.sort((a, b) -> a - b); // 1,2,3,4,5
```

```
class Person implements Comparable<Person> {
    private final String firstName, lastName;
    @Override
    public int compareTo(Person other) {
        int result = lastName.compareTo(other.lastName);
        if (result == 0)
            result = firstName.compareTo(other.firstName);
        return result;
    }
}
```

```
static int compareByAge(Person a, Person b) {
    return Integer.compare(a.getAge(), b.getAge());
}
```

```
List<Person> people = ...;
Collections.sort(people);
people.sort(Person::compareByAge);
```

COMPARATOR

```
class AgeComparator implements Comparator<Person> {
    @Override
    public int compare(Person a, Person b) {
        return Integer.compare(a.getAge(), b.getAge());
    }
}
```

```
Collections.sort(people, new AgeComparator());
people.sort(new AgeComparator());
```

```
people.sort(Comparator
    .comparing(Person::getAge)
    .thenComparing(Person::getFirstName)
    .reversed())
```

```
Comparator.comparing(Person::getName,
    (s1, s2) -> s2.compareTo(s1));
// =
Comparator.comparing(Person::getName).reversed();
```

```
Comparator<T> nullsLast(Comparator<T> c);
Comparator<T> nullsFirst(Comparator<T> c);
Comparator<T> comparing(Function<T,U> keyExtractor,
    Comparator<U> c);
Comparator<T> comparingInt(ToIntFunction<T,U> f);
```

FUNCTIONALINTERFACE

Any interface with a single abstract method is a functional interface

```
@FunctionalInterface
public interface ShortToByteFunction {
    byte applyAsByte(short s);
}

@FunctionalInterface
public interface PersonStringifier {
    String getNameAndAge(Person p);
}
```

```
COLLECTION
boolean add(E e);
boolean remove(Object o);
boolean equals(Object o);
int hashCode();
int size();
boolean isEmpty();
Object[] toArray();
void clear();
boolean contains(Object o);
boolean addAll(Collection<? extends E> c);
boolean containsAll(Collection<?> c);
boolean removeAll(Collection<?> c);
boolean retainAll(Collection<?> c);

Set<String> noDup = new HashSet<>();
```

COLLECTION IMPLEMENTATIONS

```
// List
int indexOf(Object o);
int lastIndexOf(Object o);
E get(int index);
subList(int from, int to);
void sort(Comparator<? super E> c);

// Stack
E peek();
E pop();
E push(E item);
boolean empty();
int search(Object o);

// Queue
E element(); // throws → peek(); doesn't
E remove(); // throws → poll(); doesn't
boolean add(E e); // throws → offer(E e); doesn't
```

```
// Set
// (I) SortedSet → (C) TreeSet
// (C) HashSet, (C) LinkedHashSet
```

```
// Map
// HashMap
boolean containsKey(Object key);
boolean containsValue(Object value);
Set<Map.Entry<K, V>>> entrySet();
V get(Object key);
V put(K key, V value);
V putIfAbsent(K key, V value);
V replace(K key, V value);
V remove(Object key);
V getOrDefault(Object key, V defaultValue);
Set<K> keySet();
Collection<V> values();
```

LAMBDAS

```
String pattern = readFromConsole();
// vvv not final → Error
while (pattern.length() == 0)
    pattern = readFromConsole();
Utils.removeAll(people, p ->
    p.getLastName().contains(pattern));
// Local variable ... referenced from a lambda
expression must be final or effectively final
```

```
// Predicate :: a → boolean
Predicate<Integer> isLarge = (v) -> v > 69420;
// Function :: a → b
Function<Integer, String> str = (v) -> "" + v;
// Supplier :: a
Supplier<String> hello = () -> "Hello, World!";
// Consumer :: a → void
Consumer<Integer> consomer = (v) -> log(v);
// UnaryOperator :: a → a
UnaryOperator<Integer> more = (v) -> v * v;
// BinaryOperator :: a → a → a
BinaryOperator<Integer> less = (a, b) -> a - b;
```

```
STREAMS
import java.util.stream.*;

people
    .stream()
    .distinct()
    .filter(p -> p.getAge() >= 18)
    .skip(5)
    .limit(10)
    .map(p -> p.getLastName())
    .sorted()
    .forEach(System.out::println);

people
    .stream()
    .reduce(0, (acc,cur) -> acc + cur.getAge());
```

```
list.stream().mapToInt(Integer::intValue);
list.stream().mapToInt(Integer::parseInt);
```

OPTIONAL (HASKELL / RUST IN BLOAT)

```
T get(); // NoSuchElementException
boolean isPresent();
void ifPresent(Consumer<T> consumer);
T orElse(T other);
static Optional<T> empty();
static Optional<T> of(T value);
```

METHODS

```
boolean allMatch(Predicate<T> predicate);
boolean anyMatch(Predicate<T> predicate);
boolean noneMatch(Predicate<T> predicate);
static Stream<T> concat(<Stream<T> a, Stream<T> b>);
Stream<T> distinct();
Stream<R> flatMap(<Function<T, R>> mapper);
Stream<T> limit(long maxSize);
Stream<T> skip(long maxSize);
Stream<sorted><Comparator<T> comparator>;
```

TERMINAL OPERATIONS

```
Optional<T> min(Comparator<T> comparator);
Optional<T> max(Comparator<T> comparator);
Optional<T> findAny();
Optional<T> findFirst();
void forEach(Consumer<T> action);
void forEachOrdered(Consumer<T> action);
// IntStream
average();
sum();
```

COLLECTORS

```
// List
s.collect(Collectors.toList());
// TreeSet
s.collect(Collectors.toCollection(TreeSet::new));
// String
s.collect(Collectors.joining(", "));
// Integer
s.collect(Collectors.summingInt(Person::getAge));
// Map<String, Person>
s.collect(Collectors.groupingBy(Person::getCity));
// Map<String, Integer>
s.collect(Collectors.groupingBy(Person::getCity,
    Collectors.summingInt(Person::getSalary));
// Map<boolean, List<Person>>
s.collect(Collectors.partitioningBy(s ->
    s.getAge() > 18))
```